Supplementary Material for the Article: Adaptive Scattering Transforms for Playing Technique Recognition

Changhong Wang, *Student Member, IEEE,* Emmanouil Benetos, *Senior Member, IEEE,* Vincent Lostanlen, and Elaine Chew

This document provides further details regarding datasets and detection results for our work "Adaptive Scattering Transforms for Playing Technique Recognition" [1].

A. Existing Computational Research on Playing Techniques

Besides the summary of existing computational research on playing techniques in our paper [1], we provide a complete list of playing techniques analysed and the corresponding methodologies applied in Table I.

B. Additional Information on the Datasets

Table II displays the number of ground truth frames, total duration of the ground truth, and duration of co-articulations for each playing technique in the CBFdataset. As can be seen, the number of samples for each type of playing technique is highly imbalanced. Portamento has 1010 samples while the number of trills is only 197. As compared to the duration of the whole dataset of 157 minutes the co-articulations amount to only 5 minutes. To have comparable results to the extended datasets discussed in our paper [1] and to obtain a confusion matrix between different CBF techniques, we discard the co-articulation samples and conduct a multiclass classification.

Fig. 1 shows the comparison between glissando played in isolation and those performed in full-length pieces. The isolated one consists of consecutive note changes with approximately equal duration, while the performed ones vary from each other in different ways. Player 1 lengthened the first note for the glissando at position 1, while emphasised the beginning of the glissando at position 2 by increasing the intensity. Player 2 performed the glissando together with flutter-tongue. These variabilities requires data collection of playing techniques in music context.

C. Additional Result Analysis

In Section VII-D of our paper [1], we detect all seven CBF playing techniques via a multiclass classifier using the concatenation of the proposed frequency-adaptive scattering (AdaTS+AdaTRS) and the direction-adaptive joint time–frequency scattering (dJTFS) representations as input. This is based on a performance comparison of the system using the AdaTS+AdaTRS only, the dJTFS only, and the concatenation of these two features. The concatenated feature achieves the highest marco averaged F-measure of 74.9%, as compared to

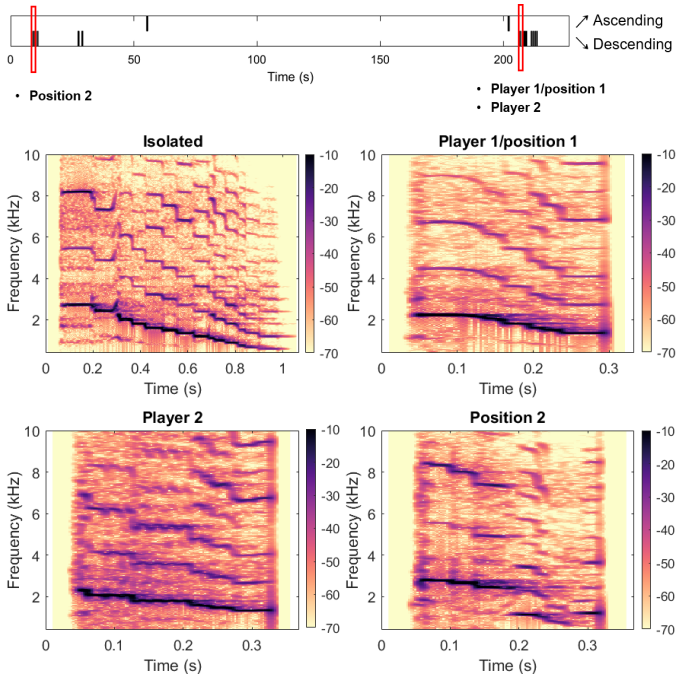

Fig. 1. Variations of glissandi between different players and different temporal positions in the performance of *Busy Delivery Harvest*.

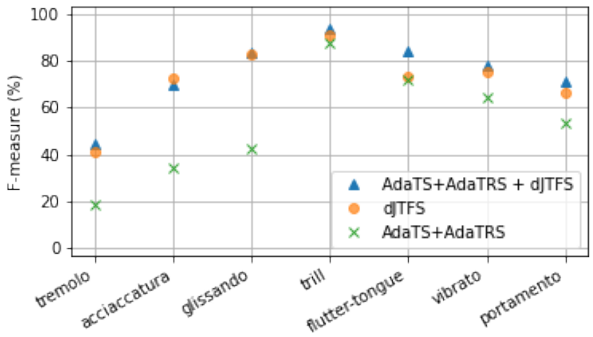

Fig. 2. Performance comparison of the recognition system on the CBFdataset using the proposed AdaTS+AdaTRS, the dJTFS, and the concatenation of AdaTS+AdaTRS and dJTFS as input.

that of 53.2% using the AdaTS+AdaTRS only and that of 71.7% using the dJTFS only. Fig. 3 shows the F-measures obtained in the multiclass classification of each playing technique using the three type of scattering representations. Although the dJTFS obtains comparable result as that of the concatenated

TABLE I
SUMMARY OF EXISTING COMPUTATIONAL RESEARCH ON PLAYING TECHNIQUES

Instrument	Expressive patterns	Methodologies applied	Cite
Guitar	Ornamentation defined by rules	Dynamic time warping sequence matching	[2]
	Legato, glissando	Symbolic, aggregate approximation	[3]
	Muted, harmonic, bend, slide, hammer-on, pull-off	Rule-based detection	[4]
	Muting, vibrato, pull-off, hammer on, sliding, bending	Sparse cepstral and phase codes	[5]–[7]
	Plucking: finger, picked, muted, slap; expression: harmonics, vibrato, bending, slide, etc.	Feature extraction, support vector machines with radial basis function kernel	[8], [9]
Piano	Trills	Independent component analysis	[10]
	Pdealling techniques	Supporter vector machines	[11]
Violin	Beginners' faults on playing techniques	K-nearest neighbours	[12]
	Flageolet, non-vibrato, pizzicato, sordino, spiccato, sul ponticello, sul tasto, tremolo	Sparse modelling	[13]
Drums	strike, buzz roll, drag, flam	Non-negative matrix factorization-based activation	[14]
	A variety of drum playing techniques	K-nearest neighbours, partial decision tree, etc.	[15]
Cello	Vibrato, tremolo, trill, glissando, etc.	Folded convolutional neural network	[16]
Irish flute	Cut, strike, crann, roll, shake	Rule-based method; hidden Markov models (HMMs)	[17]
Highland pipe	60 ornamentations	Dynamic time warping	[18]
Ney	Vibrato, kaydirma	Rule-based characterisation	[19]
Erhu	Vibrato, portamento	Filter diagonalisation method, HMMs	[20]
Guqin	None, vibrato, upward/downward/inverted-U/U protamento	Neural network	[21]
Chinese bamboo flute	Vibrato, tremolo, till, flutter-tongue, acciaccatura, portamento, glissando	HMMs, adaptive/joint time–frequency scattering	[22]–[24]
	Trill, flutter-tongue, tremolo	Synthesising	[25], [26]
Other	Vibrato, glissando	Rule-based methods	[27]
Multi-instruments	140 instrumental playing techniques (isolated)	Scattering transform	[28]
Vocal techniques	Vibrato, glissando, linear note change	COSFIRE filter	[29]
	17 different vocal techniques	VocalSet	[30]

TABLE II
NUMBER OF SAMPLES, DURATION OF GROUND TRUTH, AND DURATION OF CO-ARTICULATIONS FOR THE SEVEN TECHNIQUES IN THE CBFDATASET. TOTAL DURATION OF THE CBFDATASET IS 157 MINUTES, INCLUDING 5-MINUTE CO-ARTICULATIONS

Techniques	Flutter-tongue	Trill	Vibrato	Tremolo	Acciaccatura	Portamento	Glissando
Number of samples	628	197	519	555	441	1010	490
Duration of ground truth (minute)	7.9	15.0	13.8	6.0	1.7	6.3	5.0
Duration of co-articulations (minute)	1.3	0.1	3.7	3.7	0	0.9	0.3

feature, the latter outperforms the former for 5 out of the 7 playing techniques.

Fig. 3 and Fig. 4 are the confusion matrices obtained for recognising playing techniques in the SOL dataset and VocalSet. The recognition for the SOL dataset presents few confusions between techniques. This may be attributed to the dataset structured with short clips of individual techniques. No technique is based on the same note of the same instrument, which offers high intra-class variability for playing techniques. For the VocalSet, as can bee seen from the normalised confusion matrix in Fig. 4 (b), the lip trill obtains the best score while the trill technique obtains the lowest. There exists a large amount of confusion between trill and vibrato. The straight technique exhibits high confusion with other techniques.

Fig. 5 and Fig. 6 are the confusion matrices obtained for playing technique recognition in the CBFdataset using mel-frequency cepstral coefficients (MFCCs) and modulation power spectrum (MPS) as input features to the classifier, respectively. Both features achieve the best score for trill recognition. The MFCCs detect none of the tremolos while the MPS returns the lowest score for acciaccatura recognition. The confusion between vibrato and tremolo is also found for the recognition result by MPS.

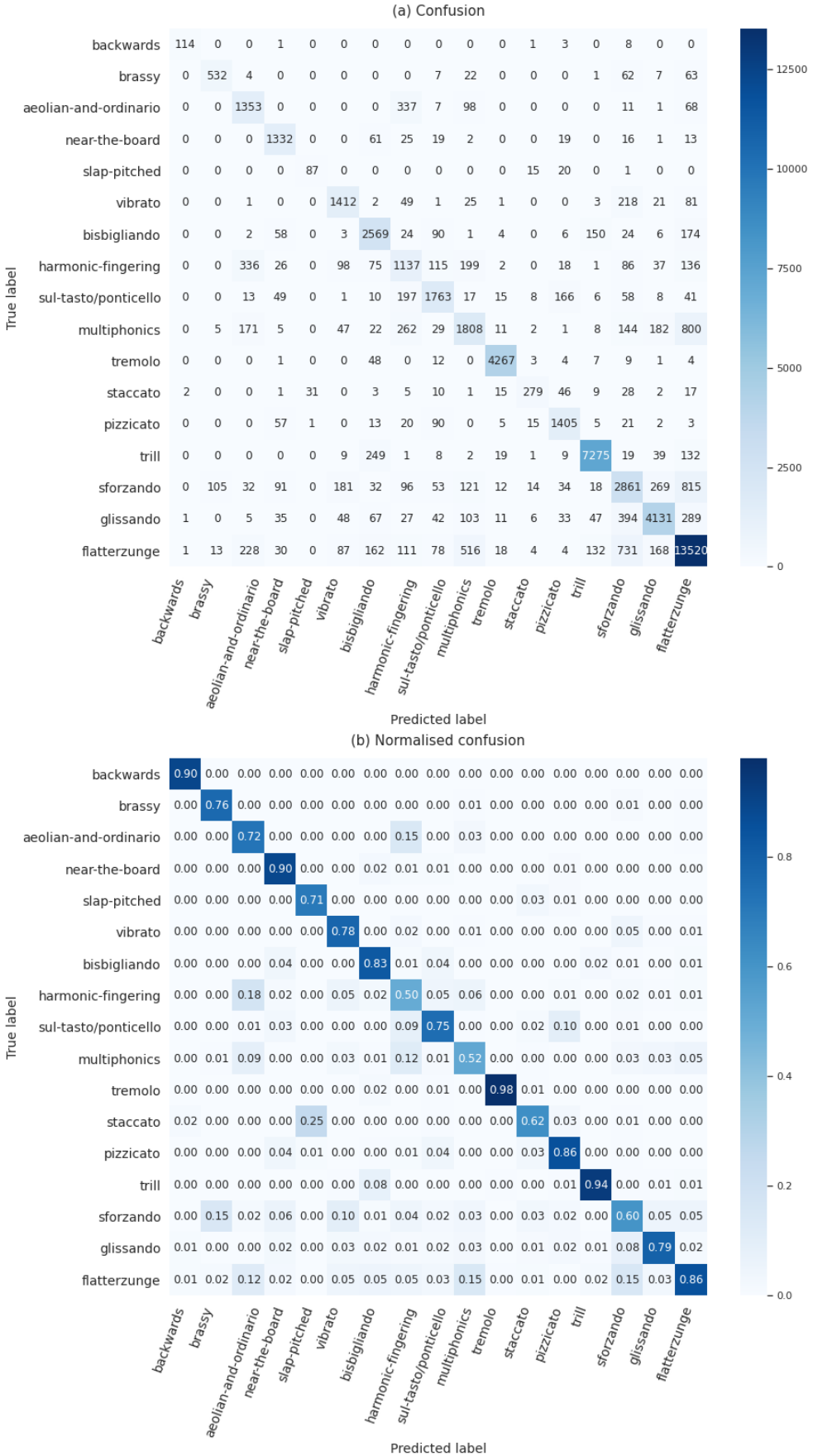

Fig. 3. Confusion matrices obtained for multi-class classification of techniques in the SOL dataset. (a): confusion matrix with number of frames detected; (b) normalised confusion matrix with (a) divided by the number of samples per technique.

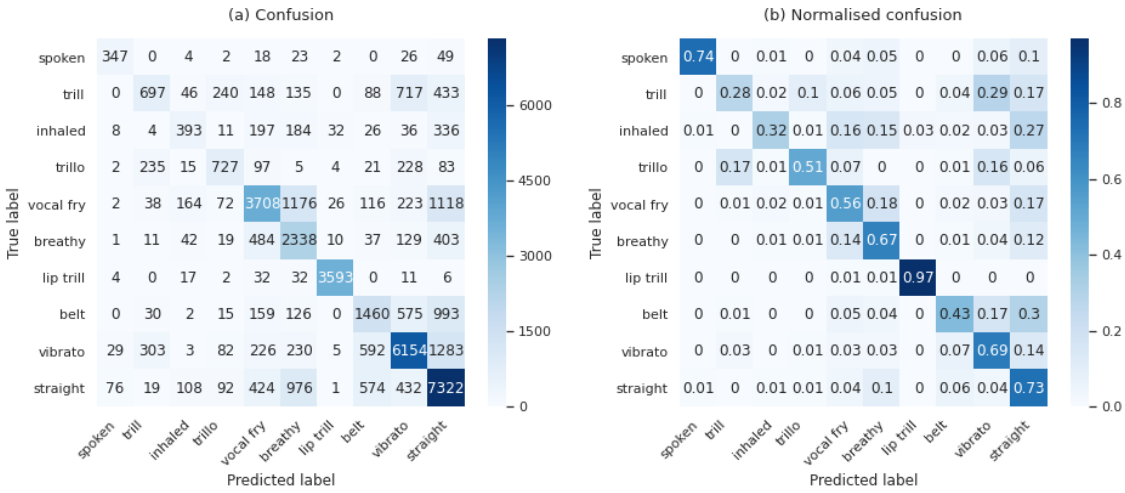

Fig. 4. Confusion matrices obtained for multi-class classification of ten vocal techniques. (a): confusion matrix with number of frames detected; (b) normalised confusion matrix with (a) divided by the number of samples per technique.

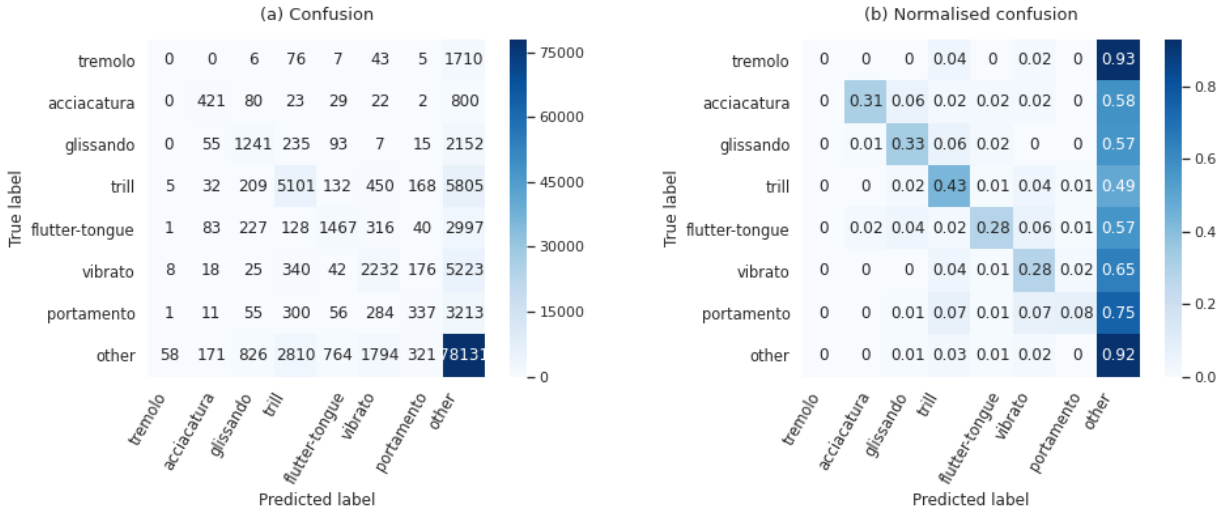

Fig. 5. Confusion matrices obtained for multi-class classification of playing techniques in the CBFdataset based on the MFCC features. (a): confusion matrix with number of frames detected; (b) normalised confusion matrix with (a) divided by the number of samples per technique.

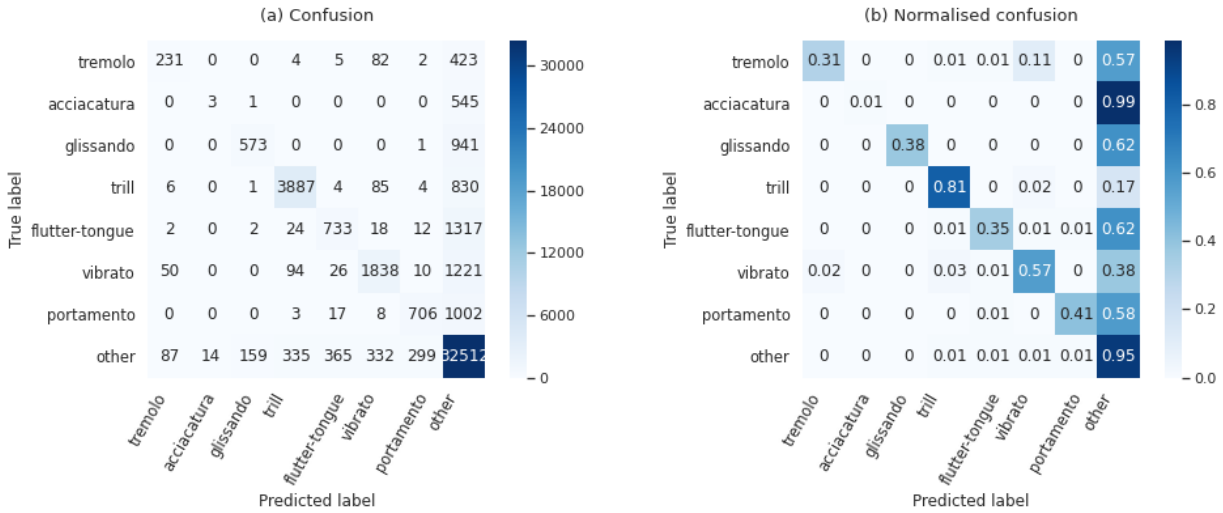

Fig. 6. Confusion matrices obtained for multi-class classification of playing techniques in the CBFdataset based on the MPS features. (a): confusion matrix with number of frames detected; (b) normalised confusion matrix with (a) divided by the number of samples per technique.

REFERENCES

[1] C. Wang, E. Benetos, V. Lostanlen, and E. Chew, "Adaptive scattering transforms for playing technique recognition," *IEEE Trans. Audio, Speech, Lang. Process.*, 2021 (submitted).

[2] S. Giraldo and R. Ramírez, "Performance to score sequence matching for automatic ornament detection in jazz music," in *Proc. Int. Conf. New Music Concepts*, Treviso, Italy, Mar. 2015.

[3] T. H. Ozaslan and J. L. Arcos, "Legato and glissando identification in classical guitar," in *Proc. Sound Music Comput. Conf*, Barcelona, Spain, Oct. 2010, pp. 457–463.

[4] L. Reboursière, O. Lähdeoja, T. Drugman, S. Dupont, C. Picard-Limpens, and N. Riche, "Left and right-hand guitar playing techniques detection." in *Proc. Int. Conf. New Interfaces Musical Expression*, Arbor, Michigan, USA, May 2012.

[5] L. Su, L. F. Yu, and Y. H. Yang, "Sparse cepstral, phase codes for guitar playing technique classification." in *Proc. Int. Soc. Music Inf. Retrieval Conf.*, 2014, pp. 9–14.

[6] P. C. Li, Y. H. Y. L. Su, and A. W. Y. Su, "Analysis of expressive musical terms in violin using score-informed and expression-based audio features," in *Proc. Int. Soc. Music Inf. Retrieval Conf.*, 2015, pp. 809–815.

[7] Y. P. Chen, L. Su, and Y. H. Yang, "Electric guitar playing technique detection in real-world recording based on F0 sequence pattern recognition," in *Proc. Int. Soc. Music Inf. Retrieval Conf.*, 2015, pp. 708–714.

[8] J. Abeßer, H. Lukashevich, and G. Schuller, "Feature-based extraction of plucking and expression styles of the electric bass guitar," in *Proc. Int. Conf. Acoust., Speech, Signal Process.*, 2010, pp. 2290–2293.

[9] J. Abeser and G. Schuller, "Instrument-centered music transcription of solo bass guitar recordings," *IEEE Trans. Audio, Speech, Lang. Process*, vol. 25, no. 9, pp. 1741–1750, 2017.

[10] J. C. Brown and P. Smaragdis, "Independent component analysis for automatic note extraction from musical trills," *J. Acoust. Soc. Am.*, vol. 115, no. 5, pp. 2295–2306, 2004.

[11] B. Liang, G. Fazekas, A. McPherson, and M. Sandler, "Piano pedaller: a measurement system for classification and visualisation of piano pedalling techniques," in *Proc. Int. Conf. New Interfaces Musical Expression*, Aalborg, Denmark, 2017, pp. 325–329.

[12] J. Charles, "Playing technique and violin timbre: Detecting bad playing," Ph.D. dissertation, Technological Univ. Dublin, Dublin, Ireland, 2010.

[13] L. Su, H. M. Lin, and Y. H. Yang, "Sparse modeling of magnitude and phase-derived spectra for playing technique classification," *IEEE Trans. Audio, Speech, Lang. Process*, vol. 22, no. 12, pp. 2122–2132, 2014.

[14] C. W. Wu and A. Lerch, "On drum playing technique detection in polyphonic mixtures." in *Proc. Int. Soc. Music Inf. Retrieval Conf.*, NY, USA, Aug. 2016, pp. 218–224.

[15] P. Herrera, A. Yeterian, and F. Gouyon, "Automatic classification of drum sounds: a comparison of feature selection methods and classification techniques," in *Int. Conf. Music Artificial Intelligence*, Edinburgh, UK, Sep. 2002, pp. 69–80.

[16] J. F. Ducher and P. Esling, "Folded cqt rcnn for real-time recognition of instrument playing techniques," in *Proc. Int. Soc. Music Inf. Retrieval Conf.*, Delft, The Netherlands, Nov. 2019, pp. 708–714.

[17] P. Jancovic, M. Köküber, and W. Baptiste, "Automatic transcription of ornamented Irish traditional flute music using hidden Markov models," in *Proc. Int. Soc. Music Inf. Retrieval Conf.*, Malaga, Spain, Oct. 2015, pp. 756–762.

[18] D. Menzies and A. McPherson, "Highland piping ornament recognition using dynamic time warping," in *Proc. Int. Conf. New Interfaces Musical Expression*, Rouge, LA, USA, May 2015, pp. 50–53.

[19] T. H. Özaslan, X. Serra, and J. L. Arcos, "Characterization of embellishments in ney performances of makam music in Turkey," in *Proc. Int. Soc. Music Inf. Retrieval Conf.*, Porto, Portugal, Oct. 2012, pp. 13–18.

[20] L. Yang, "Computational modelling and analysis of vibrato and portamento in expressive music performance," Ph.D. dissertation, Queen Mary Univ. of London, London, UK, 2017.

[21] Y. F. Huang, J. I. Liang, I. C. Wei, and L. Su, "Joint analysis of mode and playing technique in guqin performance with machine learning," in *Proc. Int. Soc. Music Inf. Retrieval Conf.*, Oct. 2020, pp. 85–92.

[22] C. Wang, E. Benetos, X. Meng, and E. Chew, "HMM-based glissando detection for recordings of Chinese bamboo flute," in *Proc. Sound Music Comput. Conf*, Malaga, Spain, May 2019, pp. 545–550.

[23] C. Wang, E. Benetos, V. Lostanlen, and E. Chew, "Adaptive Time–frequency Scattering for Periodic Modulation Recognition in Music Signals," in *Proc. Int. Soc. Music Inf. Retrieval Conf.*, Delft, The Netherlands, Nov. 2019, pp. 809–815.

[24] C. Wang, V. Lostanlen, E.Benetos, and E. Chew, "Playing technique recognition by joint time-frequency scattering," in *Proc. Int. Conf. Acoust., Speech, Signal Process.*, Barcelona, Spain, May 2020, pp. 881–885.

[25] L. Ayers, "Synthesizing trills for the Chinese dizi," in *Int. Computer Music Conf.* Singapore, 2003, pp. 227–30.

[26] ——, "Synthesizing timbre tremolos and flutter tonguing on wind instruments," in *Int. Computer Music Conf.*, 2004.

[27] R. Panda, R. Malheiro, and R. Paiva, "Novel audio features for music emotion recognition," *IEEE Trans. Affective Computing*, no. 1, pp. 1–1, 2018.

[28] V. Lostanlen, J. Andén, and M. Lagrange, "Extended playing techniques: the next milestone in musical instrument recognition," in *Proc. Int. Conf. Digital Libraries Musicology*, Paris, France, Sep. 2018, pp. 1–10.

[29] A. Neocleous, G. Azzopardi, C. N. Schizas, and N. Petkov, "Filter-based approach for ornamentation detection and recognition in singing folk music," in *Int. Conf. Computer Analysis Images Patterns*, 2015, pp. 558–569.

[30] J. Wilkins, P. Seetharaman, A. Wahl, and B. Pardo, "Vocalset: A singing voice dataset," in *Proc. Int. Soc. Music Inf. Retrieval Conf.*, Paris, France, Sep. 2018, pp. 468–474.